```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn import metrics
import seaborn as sn
import matplotlib.pyplot as plt
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving binary.dta to binary.dta
```

```
data = pd.read_stata('binary.dta')
print(data.shape)
data.head()
```

```
(400, 4)
```

| | admit | gre | gpa | rank |
|---|---|---|---|---|
| **0** | 0.0 | 380.0 | 3.61 | 3.0 |
| **1** | 1.0 | 660.0 | 3.67 | 3.0 |
| **2** | 1.0 | 800.0 | 4.00 | 1.0 |
| **3** | 1.0 | 640.0 | 3.19 | 4.0 |
| **4** | 0.0 | 520.0 | 2.93 | 4.0 |

```
data.isnull().sum()
```

| | 0 |
|---|---|
| **admit** | 0 |
| **gre** | 0 |
| **gpa** | 0 |
| **rank** | 0 |

**dtype:** int64

```
feature_cols = ['gre', 'gpa', 'rank']
x = data[feature_cols]
y = data['admit']
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=5)
display(x_train.shape, y_train.shape, x_test.shape, y_test.shape)
```

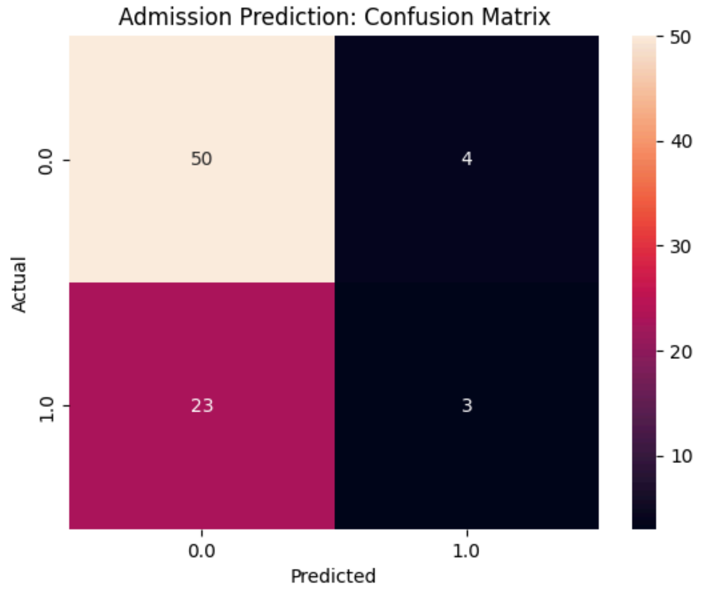
```
(320, 3)
(320,)
(80, 3)
(80,)
```

```
model = LogisticRegression(solver='lbfgs', max_iter=1000)
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
```

```
conf_mat = metrics.confusion_matrix(y_test, y_pred)
print('Confusion Matrix : ', conf_mat)
Accuracy_score = metrics.accuracy_score(y_test, y_pred)
print('Accuracy Score : ', Accuracy_score)
print('Accuracy in Percentage : ', int(Accuracy_score*100), '%')
```

```
Confusion Matrix :  [[50  4]
 [23  3]]
Accuracy Score :  0.6625
Accuracy in Percentage :  66 %
```

```
conf_mat = pd.crosstab(y_test, y_pred, rownames=['Actual'], colnames=['Predicted'])
sn.heatmap(conf_mat, annot=True, fmt='d')
plt.title('Admission Prediction: Confusion Matrix')
plt.show()
```

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